

ALEJANDRO MAYORAL

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Biomedical Engineer with experience in test engineering, manufacturing quality, incoming quality control (IQC), fixture design, and medical device development. Skilled in developing test protocols, supporting manufacturing operations, improving production processes, and implementing quality controls to increase reliability and reduce variability. Interested in production testing and quality engineering.

RELEVANT EXPERIENCE

Bioana — Product Development Engineering Intern

Monterrey, Mexico | Dic 2024 - Dic 2025

- Supported medical device development and manufacturing activities through **testing, process verification, and quality-focused engineering support**.
- **Designed and implemented incoming quality control (IQC) tests** to evaluate components, materials, and manufacturing readiness before integration into production.
- Developed **test protocols, test reports, acceptance criteria, and work instructions** to support manufacturing quality and operator execution.
- **Designed and fabricated test and manufacturing fixtures** to support verification activities, improve repeatability, and reduce process variability.
- **Executed experiments** and process evaluations in a cleanroom manufacturing environment, reducing process time by approximately **50%** in an intraocular lens production step.
- Participated in **failure analysis, PFMEA activities, and risk mitigation** initiatives to improve manufacturing robustness.
- Investigated suppliers, materials, and production equipment to support manufacturing and quality objectives.
- Collaborated with engineering and operations teams to improve manufacturing processes and define additional verification activities.

EDUCATION

Universidad de Monterrey

Monterrey, Mexico | Aug 2021 - May 2026

B.S. in biomedical engineering

Universidad Miguel Hernández

Elche, Spain | Jan 2026 - May 2026

Academic Exchange - Rehabilitation Technologies Research Project

- Designed and validated a rehabilitation assessment interface integrating a robotic system and EMG acquisition.
- Conducted structured testing and performance evaluation with users.
- Processed and analyzed test data using MATLAB and Python.
- Documented procedures, methodologies, and technical findings.

SKILLS

- Test Protocol & Report Development
- Incoming Quality Control (IQC)
- Root Cause Analysis
- PFMEA
- Fixture Design
- MATLAB, Python, Inventor, Fusion 360
- Microsoft Office: excel, word, power point